Tema 1: Circuitos Digitales

1.1 Electrónica y componentes

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Procesadores e información

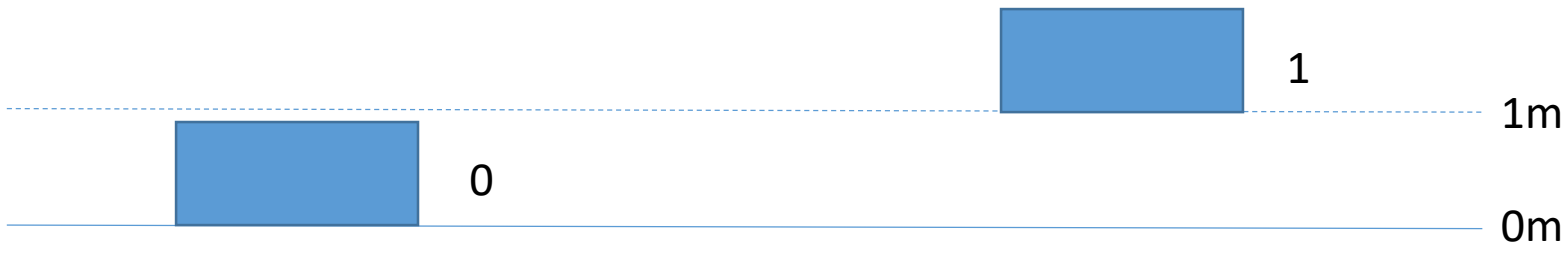
Unidad de Control

Memoria
- Instrucciones
- Datos

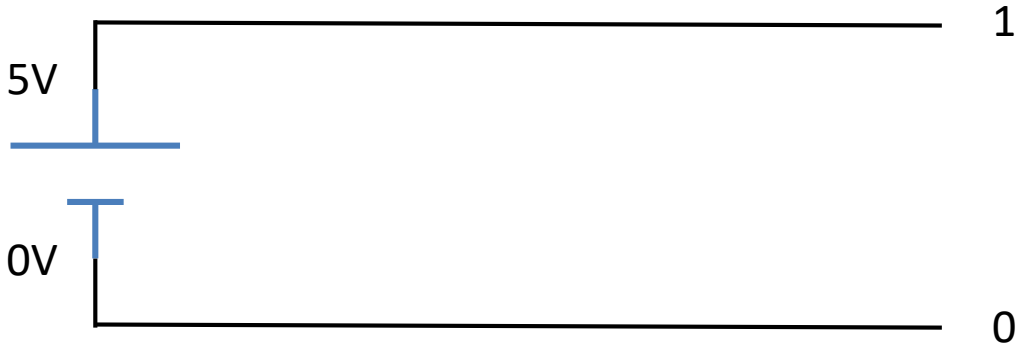
Unidad Aritmético Lógica

Almacenamiento de información

- Bit mecánico (objeto arriba o abajo)



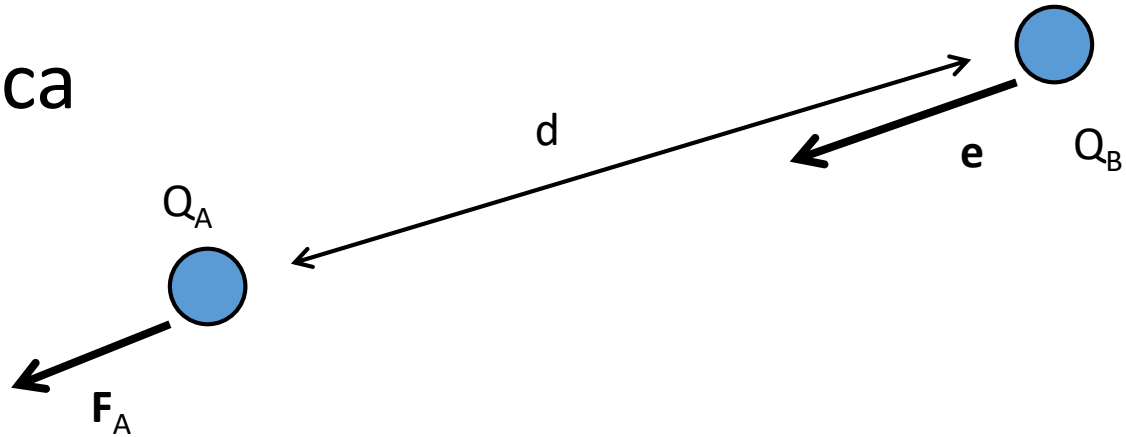
- Bit electrónico (voltaje alto o bajo)



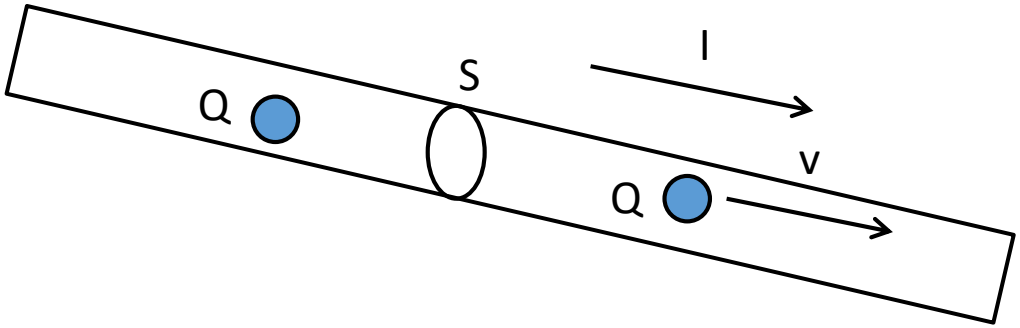
Fundamentos físicos: corriente

- Fuerza electrostática

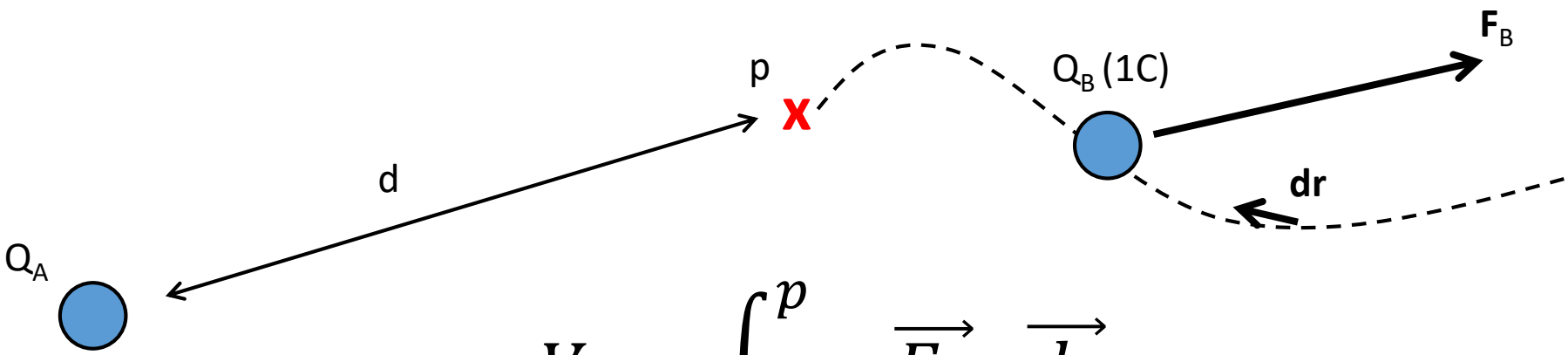
$$\overrightarrow{F_A} = k\,\frac{Q_A \cdot Q_B}{d^2}\;\vec{e}$$



- Corriente eléctrica



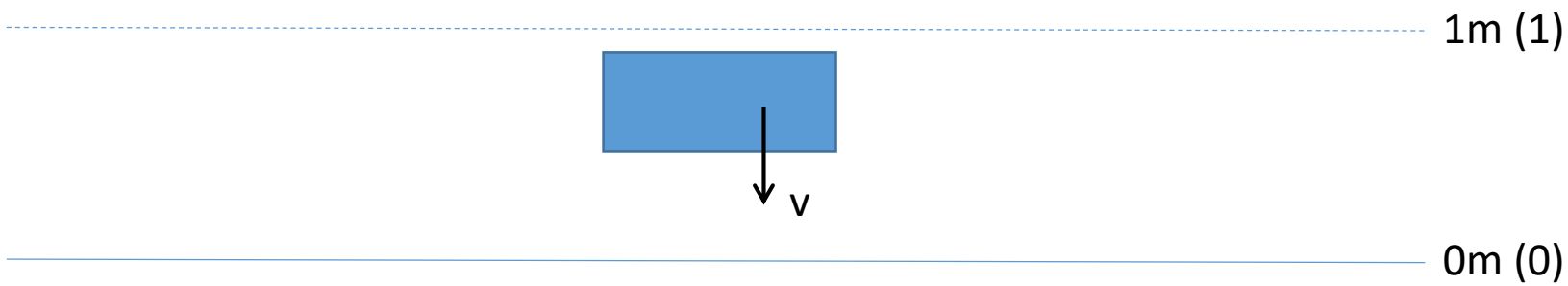
Fundamentos físicos: potencial



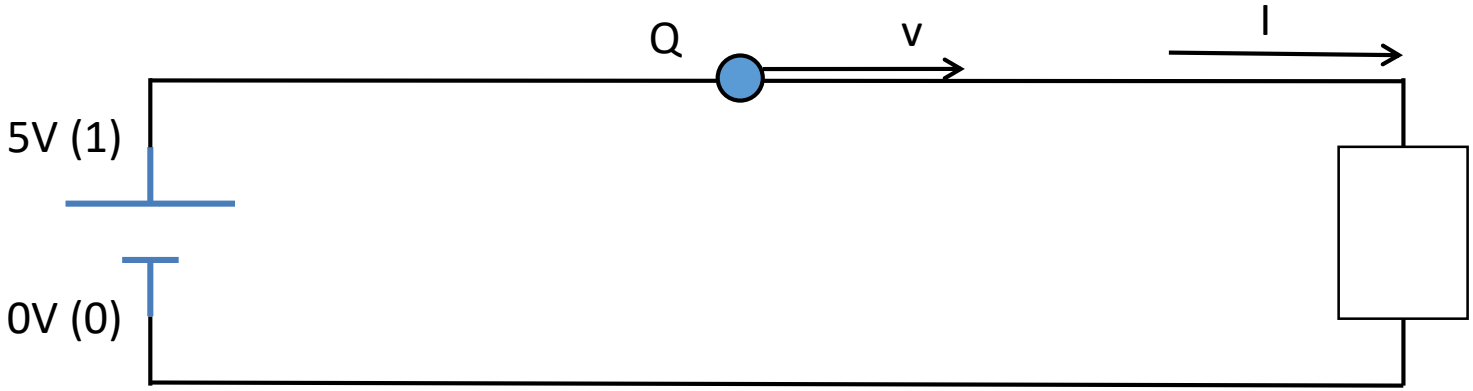
$$V_B = \int_{\infty}^{p} -\overrightarrow{F_B} \cdot \overrightarrow{dr}$$

Voltaje como estado físico

- Diferencia de alturas → Movimiento de masas

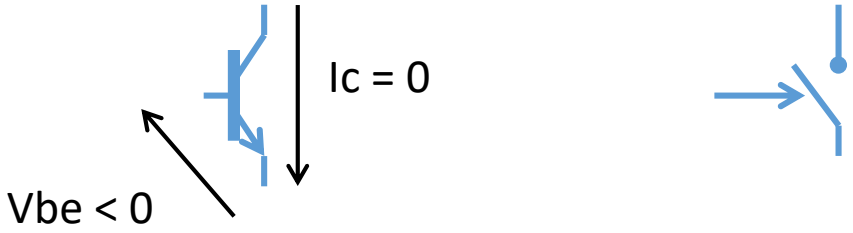


- Diferencia de potencial → Movimiento de cargas

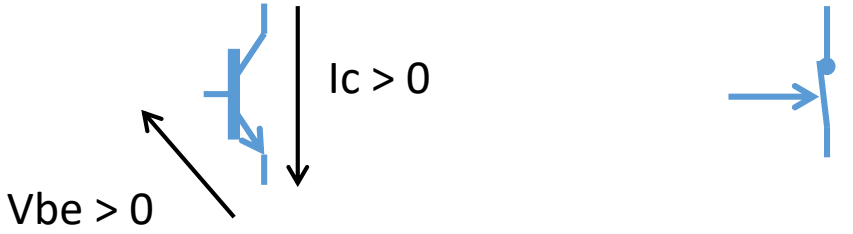


El transistor como interruptor

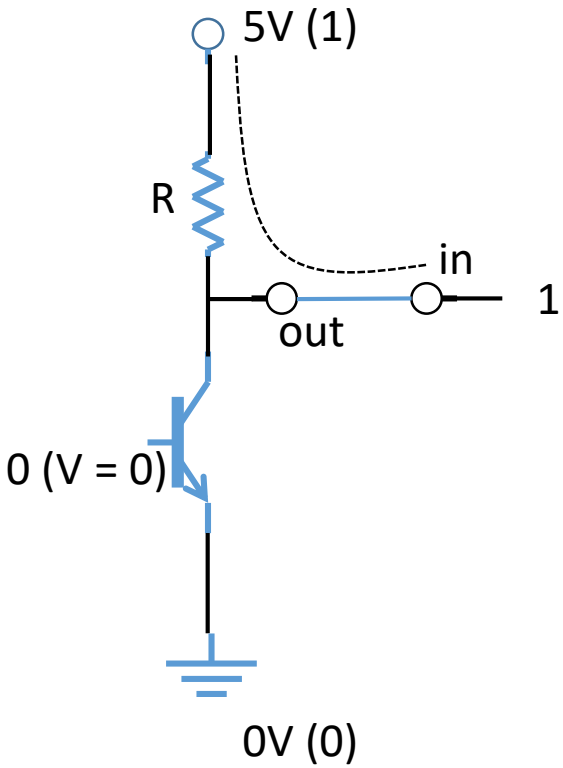
- Transistor abierto: Tensión de base 0 (o negativa); no circula corriente

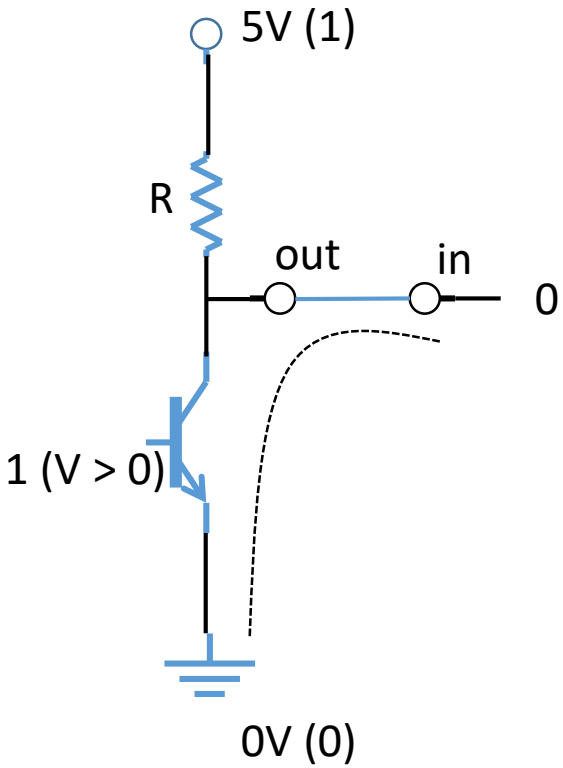
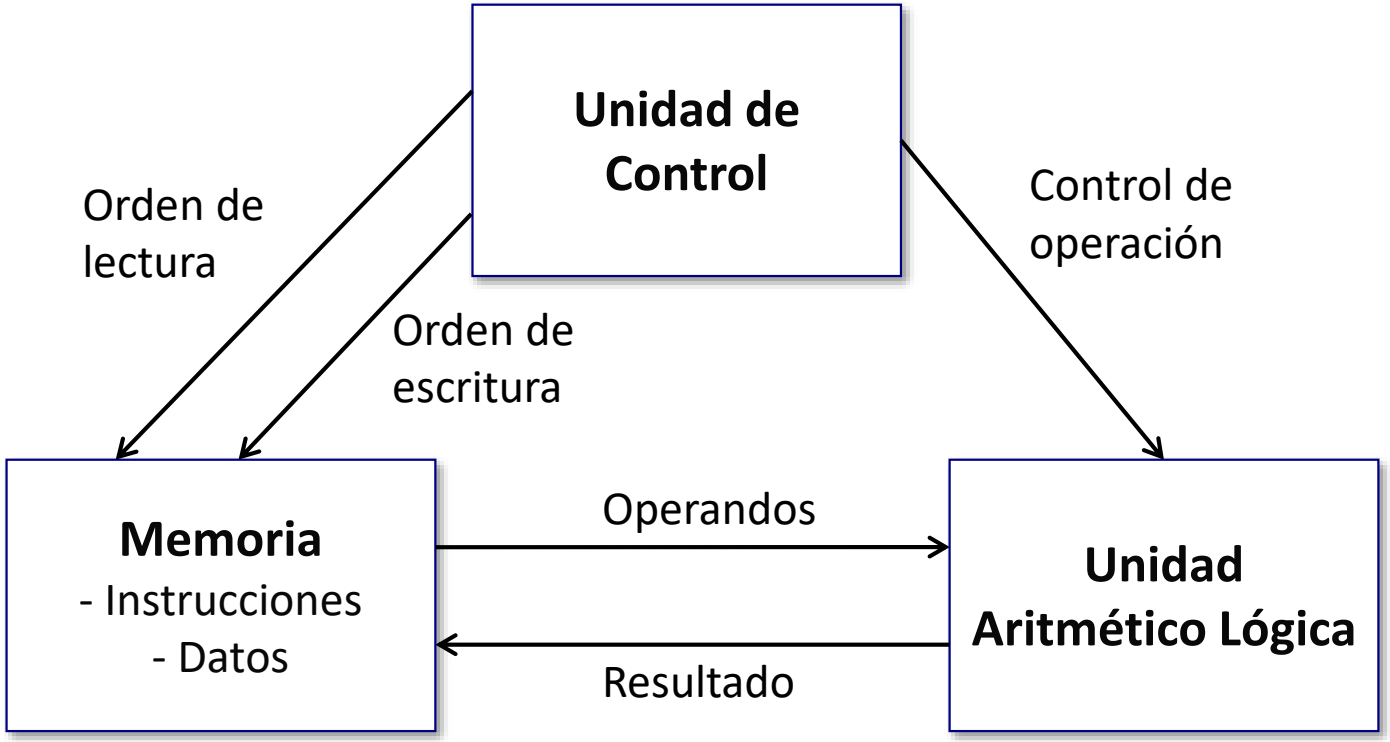


- Transistor cerrado: Tensión de base positiva; circula corriente



Interconexión de circuitos





Estudio de circuitos *bottom-up*